A Multimodal Framework for Alzheimer's Disease Classification and Stable Attribution

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Abstract

We present a multimodal framework for Alzheimer's Disease (AD) classification that combines 3D T1-weighted MRI with structured clinical text. Our 3D Vision Transformer baseline extracts neuroimaging representations that, fused with a clinical text encoder via late fusion, achieves competitive classification performance. For textual input, the Transformer Explanation Optimiser (TEO) produces the most stable feature attributions across both in-distribution and out-of-distribution cohorts, and the UMAP geometric constraint further recovers clinically consistent attributions. Interpretability analysis of the multimodal model confirms that predictions are grounded in established AD biomarkers across both modalities. These results provide insight into how stable attributions and multimodal fusion can support trustworthy clinical decision-making in AD and broader neurological applications.

1 Introduction

The application of deep learning architectures to clinical data, particularly Large Language Models (LLMs) [27] and Vision Transformers (ViTs) [5], has demonstrated significant potential in clinical neurology. These models demonstrate proficiency in diagnostic tasks, such as the classification of Alzheimer's Disease (AD) [22]. However, their integration into high-stakes clinical workflows is severely limited by their "black-box" nature.

Classical attribution-based Explainable AI (XAI) techniques, such as gradient-based approaches, perturbation methods, SHAP, and Integrated Gradients, attempt to address this by attributing importance scores to input features (e.g. tokens). Unfortunately, these methods frequently produce highly variable and unstable explanations due to entangled latent features, thereby reducing trust and clinical utility [17, 24]. A primary driver of this instability is the "superposition problem" [6], a phenomenon where neural networks compress multiple, unrelated concepts into a single dimensional activation. Consequently, classical XAI methods end up interpreting entangled, polysemantic features, leading to explanations that lack semantic clarity and inter-method consistency.

Recent advances in mechanistic interpretability address this by compressing LLM activations into a sparse, monosemantic latent space via Sparse Autoencoders (SAEs) [4]. Building directly upon the monosemantic attribution framework introduced by Mamalakis et al. [18], this project explores a unified attributional pipeline.

The main contributions of our work can be summarised as follows:

- We reproduce the textual interpretability framework for AD from [18] and analyse sparsity-stability tradeoffs.
- We develop a multimodal framework integrating textual clinical data and 3D T1-weighted MRI neuroimaging to perform a robust multimodal AD classification.
- Our attributional experiments on text baselines and the multimodal framework demonstrate how SAEs and global geometric constraints (UMAP) enhance classical XAI explanations.

2 Related Work

Attributional Explainability Classical feature attribution methods aim to distribute a model's prediction score across its input features. Gradient-based methods like Integrated Gradients and Gradient×Activation [26], alongside perturbation-based approaches like Feature Ablation and SHAP [16], have become standard practice. However, as noted by Samek et al. [24], these methods often suffer from fragility and lack of robustness, particularly in deep transformer networks where latent features are highly entangled.

Mechanistic Interpretability and SAEs Mechanistic interpretability seeks to reverse-engineer neural networks into human-understandable algorithms. Central to this field is the linear representation hypothesis [21], which posits that interpretable features correspond to fixed directions in $\mathbb{R}^d$ [20]. The superposition problem [6] actively prevents standard bases from aligning with these clear concepts. Cunningham et al. [4] demonstrated that Sparse Autoencoders (SAEs) can successfully isolate these entangled representations into highly interpretable, monosemantic features. Inspired by Mamalakis et al. [18], our methodology tests the hypothesis that extracting attribution scores from these theoretically grounded monosemantic features will yield more stable and diagnostically informative metrics than raw network activations.

Vision Transformers in Medical Imaging Vision Transformers (ViTs) have been adopted across a variety of clinical applications for medical image analysis [19, 25]. Our imaging baseline leverages these advances by utilizing a 3D ViT-B architecture pretrained via a Masked Autoencoder (MAE) on large-scale neuroimaging datasets, specifically BraTS 2023, IXI, and OASIS-3 [13]. By fine-tuning this architecture on T1 MRI neuroimaging data from the ADNI dataset, the model serves as a complementary modality to our text baseline, enabling a robust multimodal approach to Alzheimer's disease classification.

3 Methodology

The methodology of this work is structured as a hierarchical pipeline that moves from unimodal approach to a unified and interpretable multimodal framework for Alzheimer's Disease (AD) classification combining clinical text and neuroimaging modality. Our stable attribution approach is grounded in the "Monosemantic Attribution Framework" [18], which seeks to resolve the instability inherent in classical XAI by grounding feature importance in disentangled latent spaces.

3.1 Text Baseline

Data We use a dataset of clinical records from the Alzheimer's Disease Neuroimaging Initiative (ADNI) [22] and BrainLat [23] databanks. The raw clinical data is processed into a patient fact file formatted as natural language text. This text includes demographic information (age, sex, education, marital status), physical measurements (weight, pulse rate, BMI), and results from clinical assessments. The assessments cover cognitive domains including memory (Auditory Verbal Learning Test), executive function (Trail Making Test), and functional ability (Functional Activities Questionnaire). The CSV contains a binary variable for AD. For original training, we use 80% of the ADNI data for training and 20% for validation. To evaluate robustness, we use the BrainLat dataset as an out-of-distribution (OOD) test set.

Embeddings We use the ModernBERT-base [28] model to generate token-level representations. ModernBERT is a transformer-based encoder model with 24 layers and a hidden dimension of 768. We extract hidden states from the 22nd layer. We use the representation of the `[CLS]` token as the global feature vector $h_{CLS} \in \mathbb{R}^{768}$ for both downstream classification and interpretability analysis.

Monosemantic Feature Space To decompose the high-dimensional embedding space into interpretable features, we use a TopK Sparse Autoencoder (TopK-SAE) [7]. The SAE consists of an encoder and a decoder. The encoder projects the hidden states into a higher-dimensional latent space with an expansion factor of 32, resulting in $d_{SAE} = 24,576$ features. We use a Top-K activation function ($k = 64$) in the latent layer to enforce hard sparsity. This ensures that only the K most active features are non-zero for any given input. Before processing, we normalise the input activations by

their mean Euclidean norm. This normalisation ensures that the reconstruction error and sparsity penalties remain scale-invariant across layers and models, which stabilises training.

Attribution Optimisation We use the Transformer Explanation Optimiser (TEO), an encoder-decoder architecture based on 1D transformers, to generate stable and sparse attribution maps. The model is trained to minimise a multi-objective cost function that balances stability, sparsity, and similarity to a consensus target. The consensus target $\bar{\phi}$ is a weighted average of attributions from six methods: Gradient Activation, Integrated Gradients, Layer Conductance, Gradient SHAP, Feature Ablation, and raw Activations.

For the TEO and TEO-SAE variants, the loss function L_{base} is:

$$L_{base} = \lambda_1 R_{RIS}^{-1} + \lambda_2 R_{ROS}^{-1} + \lambda_3 M_{sparse} + \lambda_4 L_{similarity} \tag{1}$$

where R_{RIS}^{-1} and R_{ROS}^{-1} are inverse stability proxies, M_{sparse} is an L_1 sparsity penalty, and $L_{similarity}$ is the mean squared error relative to the consensus target.

For the TEO-UMAP variant, we introduce a geometry-aware constraint ($\lambda_5 L_{umap}$) to impose global linear structure on the explanation space. This constraint requires equality between the first two principal components of the attribution manifold, effectively regularising the explanations so that dominant patterns are consistent across the cohort. The total loss function is:

$$L_{total} = L_{base} + \lambda_5 L_{umap} \tag{2}$$

The parameters λ are set to (0.1, 0.3, 0.1, 0.5) for the base terms, with $\lambda_5 = 0.1$ for the UMAP variant.

Evaluation We evaluate the interpretability of our model using the Quantus framework [9]. We measure sparseness [3] using the Gini index and evaluate the stability of attributions using Relative Input Stability (RIS) and Relative Output Stability (ROS) [2]. These metrics are calculated by perturbing the input features and measuring the consistency of the resulting importance scores. Performance is validated across both in-distribution (ADNI) and out-of-distribution (BrainLat) datasets to ensure the generalisability of the explanations. We also perform further investigations on the trade-off between stability and sparsity using PCA projections and cumulative variance analysis.

3.2 Neuroimaging Baseline

Data We use T1-weighted MRI scans from the ADNI dataset [22]. The raw volumes are provided as pre-processed NIfTI files that have undergone spatial normalisation to MNI space, skull stripping, and N3 bias field correction through the ADNI processing pipeline. All volumes share a uniform shape of $110 \times 110 \times 110$ voxels at 2mm isotropic resolution. We apply three additional preprocessing steps: (1) reorientation from LAS to RAS convention using the NIfTI metadata, (2) per-volume z-score intensity normalisation, and (3) resizing to $128 \times 128 \times 128$ to match the ViT's input requirements. The resulting dataset has 214 labelled subjects, partitioned as shown in Table 1.

Split	**Total**	**AD**	**CN**	**AD (%)**
Train	84	44	40	52.4
Validation	22	11	11	50.0
Test	108	66	42	61.1
Total	214	121	93	56.5

Table 1: T1 neuroimaging dataset split and label distribution. The training set is approximately balanced, while the test set is skewed toward AD (61.1%).

Architecture and Training As the image baseline, we use the 3D ViT-B MAE architecture from Kunanbayev et al. [13], a volumetric Vision Transformer pre-trained with a Masked Autoencoder

(MAE) objective [8] on large-scale neuroimaging data (BraTS 2023 [1], IXI [10], and OASIS-3 [14]) using a 75% masking ratio. The encoder consists of 12 transformer blocks with a hidden dimension of 768 and 12 attention heads. The input 128^3 volume is partitioned into non-overlapping 16^3 patches, yielding $8 \times 8 \times 8 = 512$ patch tokens. A learnable `[CLS]` token is prepended to the sequence, and positional embeddings are added. This pre-trained initialisation is particularly important in our setting, as the available T1 training set (84 subjects) is very small for a high-capacity 3D transformer with 88.6M parameters.

For AD classification, the `[CLS]` token output from the final transformer layer provides a 768-dimensional global representation, which is mapped to the binary AD/CN prediction by a linear classification head ($\mathbb{R}^{768} \rightarrow \mathbb{R}^2$). We fine-tune the full model using AdamW [12] with a learning rate of 10^{-4}, weight decay of 0.3, and a cosine annealing schedule [15] preceded by a 10-epoch linear warmup. A weighted cross-entropy loss [11] accounts for the class imbalance. Early stopping with patience of 15 epochs on validation AUROC is used to select the best checkpoint.

This baseline serves two purposes. First, it establishes how much predictive signal can be extracted from T1 MRI alone. Second, it provides the visual encoder for the multimodal model, where its frozen `[CLS]` feature acts as the neuroimaging representation.

3.3 Multimodal Framework

To construct the multimodal classifier, we matched the T1 cohort to the clinical-text cohort, referred to as matched cohort, using patient identifiers and used the T1 split (Table 1) as the data split. All 214 T1 subjects had matching clinical text records, producing a subject-level matched cohort with the same 106 train set and 108 held-out test subjects used in the neuroimaging baseline.

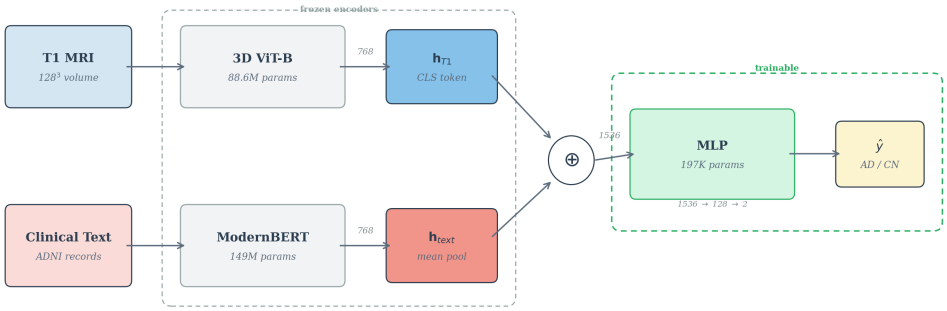


Figure 1: Multimodal architecture used for AD/CN classification from T1 MRI and clinical text. A frozen 3D ViT-B encoder produces a 768-dimensional CLS representation from the preprocessed T1 volume, while a frozen ModernBERT encoder produces a 768-dimensional mean-pooled text representation. The two embeddings are concatenated into a 1536-dimensional fused feature and passed to a lightweight trainable MLP head ($1536 \rightarrow 128 \rightarrow 2$) for final prediction.

Embedding Extraction For each patient, we extract a 768-dimensional feature vector from each modality. The *vision embedding* $\mathbf{h}_{\text{T1}} \in \mathbb{R}^{768}$ is the `[CLS]` token output from the fine-tuned ViT-B encoder. The *text embedding* $\mathbf{h}_{\text{text}} \in \mathbb{R}^{768}$ is the mean-pooled output of the last hidden layer (layer 22) of the fine-tuned ModernBERT model, computed over non-padding tokens:

$$\mathbf{h}_{\text{text}} = \frac{\sum_{i=1}^{L} m_i \cdot \mathbf{h}_i^{(22)}}{\sum_{i=1}^{L} m_i} \tag{3}$$

where $\mathbf{h}_i^{(22)}$ is the hidden state at position i of layer 22 and $m_i \in \{0, 1\}$ is the attention mask. This pooling strategy matches the `classifier_pooling="mean"` configuration used during ModernBERT fine-tuning. Both encoders remain frozen throughout the training; only the parameters in the fusion architecture are updated.

Fusion Architecture The two embeddings are concatenated into a 1536-dimensional fused representation:

$$\mathbf{z} = [\mathbf{h}_{\text{T1}}; \mathbf{h}_{\text{text}}] \in \mathbb{R}^{1536} \tag{4}$$

The fused representation is passed to a lightweight multilayer perceptron:

$$\hat{y} = \text{MLP}(\mathbf{z}) \tag{5}$$

where the MLP consists of Linear$(1536, 128) \rightarrow$ GELU $\rightarrow$ Dropout$(0.3) \rightarrow$ Linear$(128, 2)$. This head contains approximately 197K trainable parameters, while both encoders remain frozen during fusion training.

We chose this late-fusion design because the matched multimodal cohort is extremely small. Under such conditions, freezing both encoders reduces overfitting risk and allows the model to exploit already-learned unimodal representations. A more complex fusion mechanism such as token-level cross-attention was considered unnecessary given the available data.

Training Protocol Training proceeds in two phases. In *Phase A*, we merge the training and validation subjects into a 106-subject set and perform 5-fold cross-validation. Each fold is trained with AdamW (lr $= 10^{-3}$, weight decay $= 0.1$), cosine annealing, weighted cross-entropy loss with label smoothing ($\epsilon = 0.1$), and early stopping with patience of 30 epochs on fold-validation AUROC. This phase is used for model selection, threshold selection. In *Phase B*, we train once on all 106 subjects for a fixed number of epochs (the average best epoch from Phase A) and evaluate once on the 108 held-out test subjects. The classification threshold is locked to the median threshold from Phase A to prevent data leakage. To quantify optimisation variance, the final multimodal model was repeated across five runs and we report mean $\pm$ standard deviation on the held-out test AUROC.

3.4 Multimodal Interpretability

To understand how the multimodal model integrates information from images and text, we apply three gradient-based attribution methods to a subset of four correctly classified test patients (two AD, two CN). Crucially, all gradients originate from the *fusion model's* predicted-class logit, not from the standalone unimodal classifiers, ensuring that the attributions reflect the multimodal decision.

Modality Contribution We compute Gradient$\times$Input [26] on the 1536-dimensional fused input vector. For each patient, the gradient of the predicted-class logit is computed with respect to the concatenated embedding $\mathbf{z}$. The per-modality attribution is:

$$A_v = \sum_{d=1}^{768} \left| \frac{\partial \hat{y}_c}{\partial z_d} \cdot z_d \right|, \quad A_t = \sum_{d=769}^{1536} \left| \frac{\partial \hat{y}_c}{\partial z_d} \cdot z_d \right| \tag{6}$$

where $\hat{y}_c$ is the logit for the predicted class c. The relative contribution of each modality is then $A_v/(A_v + A_t)$ and $A_t/(A_v + A_t)$.

Brain Spatial Importance To identify which brain regions contribute to the prediction, we employ Gradient$\times$Activation on the ViT's internal patch token representations. Let $\mathbf{h}_i \in \mathbb{R}^{768}$ denote the activation of patch token i at the second-to-last transformer layer, where $i \in \{1, \ldots, 512\}$. The importance score for each patch is:

$$s_i = \left\| \frac{\partial \hat{y}_c}{\partial \mathbf{h}_i} \odot \mathbf{h}_i \right\|_1 \tag{7}$$

The resulting 512-dimensional importance vector is reshaped into an $8 \times 8 \times 8$ spatial grid and upsampled to 128^3 for visualisation as a heatmap overlaid on the original T1 volume. Since the model is patch-based, these maps should be interpreted as coarse patch-level importance rather than precise voxel-level localisation.

Text Token Importance To identify which clinical features in the text drive the prediction, we compute gradients from the fusion logit through the frozen ModernBERT encoder back to the token-level input embeddings. The gradient flows from the fusion MLP through the mean-pooling operation and all 22 ModernBERT layers. The importance of each token j is:

$$r_j = \left\| \frac{\partial \hat{y}_c}{\partial \mathbf{e}_j} \odot \mathbf{e}_j \right\|_2 \tag{8}$$

where $\mathbf{e}_j \in \mathbb{R}^{768}$ is the input embedding at position j. The resulting importance scores are mapped back to the original clinical text, allowing direct visualization of which words and numerical values contribute most to the multimodal prediction.

4 Results

4.1 Text Baseline

Dataset	Accuracy	F1 Score	AUROC	AP
ADNI test data	0.769	0.768	0.841	0.866
BrainLAT (OOD)	0.490	0.641	0.447	0.467

Table 2: Binary classification performance metrics across ADNI and BrainLAT cohorts. Metrics include Accuracy, F1 Score (harmonic mean of precision and recall), AUROC (Area Under the Receiver Operating Characteristic curve), and AP (Average Precision).

Attribution Method	Sparsity (↑)	RIS (↓)	ROS (↓)
ADNI Test Set Binary Classification			
Grad Activation	0.5833	3.8105	2204.5065
Grad Activation-SAE	0.6409	27.7976	20352.4535
Integrated Gradients	0.5843	1.6008	1693.4563
Integrated Gradients-SAE	0.6324	2.5701	11036.7876
Layer Conductance	0.5843	2.2082	1767.3199
Layer Conductance-SAE	0.6324	2.4928	12073.4949
Gradient SHAP	0.5827	538.9932	71083.4956
Gradient SHAP-SAE	0.6332	361.6156	185731.6552
Feature Ablation	0.5823	2.3927	2010.3582
Feature Ablation-SAE	**0.7121**	1593.2538	1365654.3191
Activations	0.4643	1.0000	1806.1203
TEO	0.4918	**0.0014**	**0.3681**
TEO-SAE	0.4844	0.0029	3.0094
TEO-UMAP	0.4858	0.0031	3.6815
BrainLAT (OOD) Test Set Binary Classification			
Grad Activation	0.5748	1.5087	6674.3445
Grad Activation-SAE	0.5967	15.5834	35894.3023
Integrated Gradients	0.5739	3.5290	7728.2270
Integrated Gradients-SAE	0.6147	5.2083	49793.0662
Layer Conductance	0.5739	2.6684	7351.8328
Layer Conductance-SAE	0.6147	4.3427	47173.9410
Gradient SHAP	0.5747	155.3349	82486.4191
Gradient SHAP-SAE	0.6113	108.4743	279769.6896
Feature Ablation	0.5758	2.0251	9163.9911
Feature Ablation-SAE	**0.6956**	741.7387	766623.5887
Activations	0.4586	1.0000	7144.7905
TEO	0.4825	0.0038	**2.4044**
TEO-SAE	0.4798	**0.0035**	12.6783
TEO-UMAP	0.4812	0.0047	14.0436

Table 3: Performance metrics for ADNI and BrainLAT cohorts. Arrows indicate the preferred direction for each metric: higher Sparsity (↑), lower Relative Importance Score (RIS, ↓), and lower Relative Output Sensitivity (ROS, ↓). TEO variants achieve the lowest RIS and ROS across both cohorts. Bold indicates optimal column results.

Classification Performance The classification results on ADNI and BrainLAT (Table 2) match those reported in [18]. This confirms that the model learns discriminative clinical patterns from

structured text. In-distribution performance on ADNI is strong. Performance drops on BrainLAT due to distinct clinical environments: the cohort originates from Latin American multi-site collection protocols, with different demographic profiles, neuroimaging acquisition procedures, and cognitive assessment conventions compared to ADNI. Generalisation across these datasets is difficult; however, the F1 score of 64.1% demonstrates that clinical signals remain partially detectable under zero-shot transfer.

Attribution Stability and Sparsity The attribution results (Table 3) show that TEO variants outperform all baselines in RIS and ROS. Lower RIS/ROS indicates that attributions remain consistent under small perturbations to the hidden-state input which is essential for trustworthy explanations.

Sparseness increases for SAE variants across most configurations, as expected. The SAE encourages representations with a small number of active monosemantic features for any given input. This concentrates attribution mass. For most methods, RIS and ROS also increase when the SAE is introduced, which indicates a stability/sparsity trade-off. The SAE maps the LLM's polysemantic activations into a higher-dimensional but sparser feature space. When classical attribution methods operate in this space, small perturbations to the input can shift the sparse activation pattern discontinuously, and this decreases stability.

TEO's multi-objective loss resolves this trade-off by penalising RIS and ROS during training. TEO-SAE achieves the lowest RIS and ROS values while retaining sparseness. The sparseness stabilises to values near 0.49 on ADNI. The linear constraint in TEO-UMAP restores sparseness at a modest stability cost relative to TEO-SAE. This offers a tunable point depending on whether selectivity or stability is the primary requirement.

The Gradient Activation baseline yields a notably higher ROS on BrainLAT than the values of 16–20 reported in [18]. Perhaps this discrepancy is due to where the metric is measured. In our implementation, RIS and ROS are computed on the layer-22 CLS embedding with a 10% perturbation bound, rather than on raw input tokens. Gradient$\times$Activation amplifies high-magnitude gradient regions, and at the hidden-state level these magnitudes are considerably larger than at the input-embedding level, which drives ROS upward. The original paper does not explicitly specify the measurement point, and this prevents a direct comparison.

OOD Generalisation Although TEO is optimised exclusively on IID ADNI explanations, the attribution metrics show decent performance on the OOD BrainLAT data. The UMAP linear constraint adds population-level regularisation that encourages the explanation space to reflect structural properties of the clinical domain rather than dataset-specific artefacts.

Geometry of the Explanation Space The PCA projections (Figures 3 and 4) demonstrate that the attribution space for TEO and its variants is less concentrated than standard methods. Baseline methods appear as simple linear structures in the first two principal components, which indicates that attributions collapse onto a single dominant direction. This reflects the polysemantic nature of the underlying representations: when multiple concepts are entangled in the same feature dimensions, the attribution signal reduces to a single axis of variation. TEO-based explanations form complex, distributed point clouds in PCA space. The clinical categories are spread across multiple directions rather than collapsed onto one.

The cumulative variance analysis (Figure 2) confirms this quantitatively. Baseline methods concentrate the majority of their explained variance in the first one or two components. TEO spreads its mass across more principal components, which means that clinical information is distributed across a higher-dimensional manifold. TEO achieves stability without collapsing the explanation onto a few directions. Instead, it preserves a richer geometric structure across the cohort.

Individual Patient Attribution We investigate the attribution pattern for Patient 4501, a Male AD case (Figure 5). Attributions are thresholded at 15% of the maximum absolute value per method; green indicates evidence for AD, red indicates evidence against AD.

TEO assigns predominantly negative attributions to this patient despite the positive ground truth. The strongest red activations concentrate on demographic tokens: *sex*, *Male*, *Right* (handedness), and *Not Hispanic*. Within the cognitive assessments, TEO highlights CDT score labels (*Clock*, *Total Score*), AVLT trial indices (6, 5, 4), and intrusion counts with red attributions. This suggests the polysemantic

target encodes a population-level prior: males have a lower base rate of AD than females in the ADNI cohort. Therefore, the gradient signal is steered by this demographic imbalance. Individual clinical evidence that contradicts the demographic prior is suppressed.

TEO-SAE produces a sparser but also negative attribution for this patient. The SAE bottleneck promotes monosemantic features; however, the active features still encode the Male sex signal (*Male* in red), and most CDT and AVLT outcome tokens remain negative. Two positive signals emerge, *education* and the blood pressure fields (*Systolic mmHg*, *Diastolic mmHg*). This indicates that the monosemantic decomposition begins to isolate physiological risk features.

TEO-UMAP reverses the dominant polarity: *Male* now receives a positive attribution, and CDT tokens as well as AVLT tokens are predominantly green. The UMAP linear constraint forces the learned explanation space to preserve the geometric structure of the SAE feature manifold across the training population. This discourages the model from collapsing onto a single demographic axis and encourages it to weight cognitive assessment outcomes directly.

The disagreement between TEO and TEO-UMAP on the same patient is informative. TEO operates in the polysemantic layer-22 space, where sex and cognitive decline co-vary. The attribution reflects this rather than the causal pathway. TEO-UMAP, through population-level geometric regularisation, decorrelates these directions and recovers attributions consistent with the clinical record.

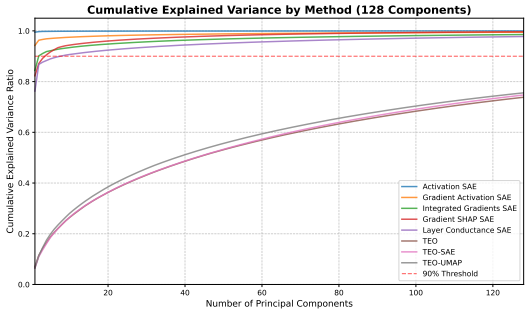


Figure 2: Cumulative explained variance. TEO's attribution space is higher-dimensional and more spread out than standard methods.

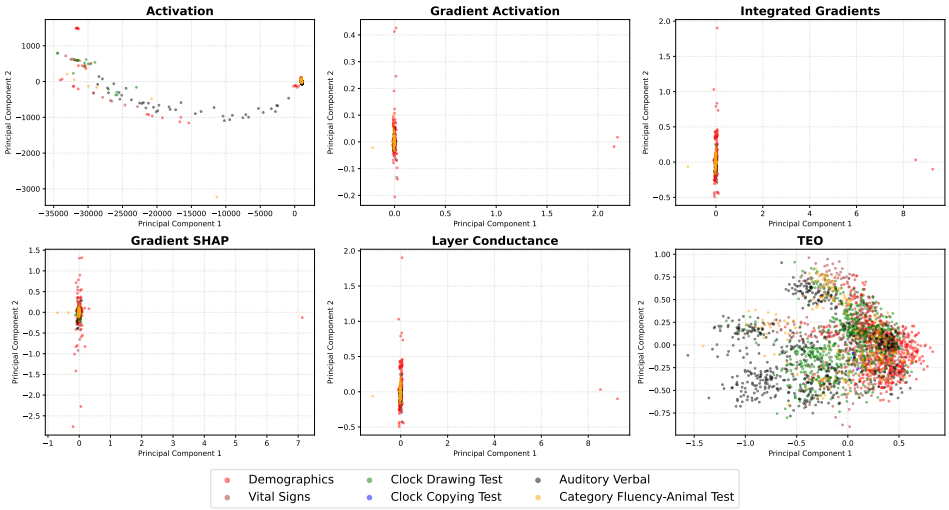


Figure 3: PCA projections of attribution vectors for non-SAE methods. Baselines appear as simple 'straight lines' dominated by single features.

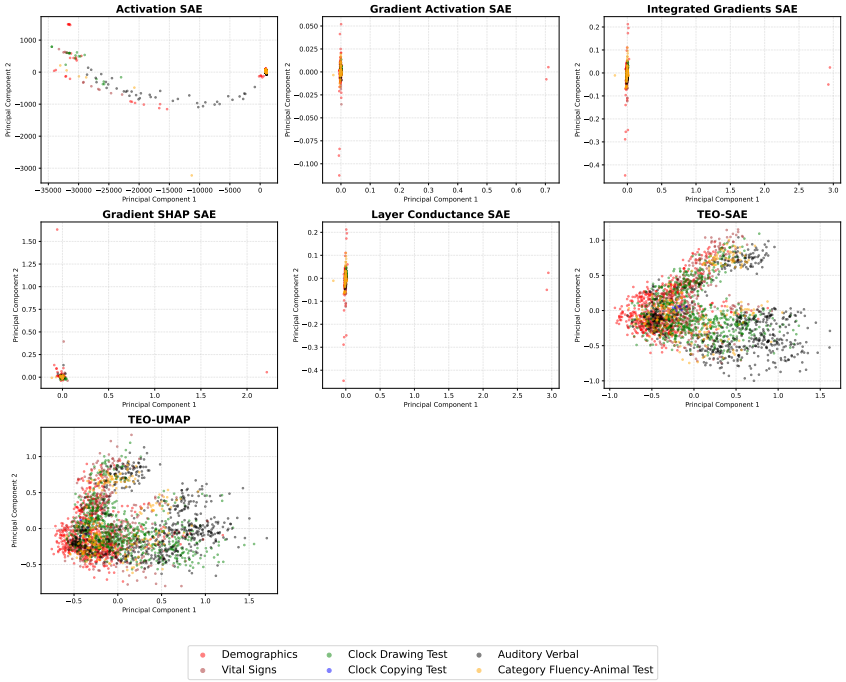


Figure 4: PCA projections of attribution vectors in the SAE feature space. TEO variants produce complex point clouds while baselines remain dominated by 1D structures.

Figure 5: Patient activations for patient 4501 with confirmed AD across TEO, TEO-SAE, and TEO-UMAP. Green represents tokens which indicate AD (positive result) whereas red indicates control (negative result).

4.2 Neuroimaging Baseline

AUROC	$0.64 \pm .05$
Balanced Accuracy	0.612
F1 Score	0.587
Sensitivity	0.485
Specificity	0.738

Table 4: Neuroimaging baseline performance on the ADNI test set ($n = 108$). AUROC is mean $\pm$ std across 5 runs.

The neuroimaging baseline achieves a test AUROC of 0.64 ± 0.05 across five independent runs (Table 4). The model exhibits moderate specificity (0.738) but low sensitivity (0.485), indicating difficulty in detecting AD cases from structural MRI alone at this dataset size. Overfitting is observed within 10–15 epochs, with training loss approaching zero while validation loss increases. Despite these limitations, the learned representations prove valuable when combined with text features in the multimodal setting.

4.3 Multimodal Framework

Across five repeated runs on the held-out test set, the multimodal model achieved an AUROC of 0.820 ± 0.009, demonstrating both competitive performance and low optimisation variance.

The fairest comparison is against unimodal probes trained on the same limited matched-data regime, namely the 106-subject multimodal cohort. Table 5 presents the full evaluation metrics under this setting.

Model	**AUROC**	**Balanced Accuracy**	**F1**	**Sensitivity**	**Specificity**
T1 MRI (unimodal probe)	$0.665_{\pm.009}$	$0.613_{\pm.002}$	$0.610_{\pm.003}$	$0.521_{\pm.007}$	$0.705_{\pm.012}$
Clinical text (unimodal probe)	$0.782_{\pm.011}$	$0.711_{\pm.015}$	$\mathbf{0.761}_{\pm.012}$	$\mathbf{0.736}_{\pm.015}$	$0.686_{\pm.028}$
Multimodal model	$\mathbf{0.820}_{\pm.009}$	$\mathbf{0.742}_{\pm.012}$	$0.759_{\pm.012}$	$0.694_{\pm.022}$	$\mathbf{0.790}_{\pm.028}$

Table 5: Fair same-data comparison on the matched multimodal cohort ($n = 108$ test). All models were trained on the same 106-subject development set with frozen encoder embeddings.

The multimodal model achieves the highest AUROC (0.820 ± 0.009) and balanced accuracy (0.742 ± 0.012) among the three models. While the text probe attains a higher F1 score (0.761 ± 0.012) and sensitivity (0.736 ± 0.015), the fusion model achieves a better balance between sensitivity (0.694 ± 0.022) and specificity (0.790 ± 0.028) the text probe exhibits higher sensitivity but lower specificity (0.686 ± 0.028), while the neuroimaging probe shows the opposite pattern. This suggests that the two modalities provide complementary predictive signals that, when combined, yield a more balanced classifier.

For context, Table 6 compares the multimodal model against all baselines, including the original unimodal models trained under their own (different) data regimes, referred to as original head. The multimodal model is competitive with the original fine-tuned ModernBERT classifier, but does not clearly outperform it. This is an expected outcome: the text model benefited from a richer text-only training dataset and therefore remains the strongest single modality overall.

Model	AUROC	F1	Training Regime
T1 MRI (original head)	0.700	0.587	84 T1 scans
ModernBERT (original head)	0.833	0.768	1142 text subjects
T1 MRI (unimodal probe)	$0.665_{\pm.009}$	$0.610_{\pm.003}$	matched cohort
Clinical text (unimodal probe)	$0.782_{\pm.011}$	$\mathbf{0.761}_{\pm.012}$	matched cohort
Multimodal model	$\mathbf{0.820}_{\pm.009}$	$0.759_{\pm.012}$	matched cohort

Table 6: Full comparison across all baselines. The top two rows use the original unimodal classification heads trained under different regimes; the bottom three rows are trained on the same 106-subject matched cohort (fairest comparison). Bold indicates best out of matched cohort results.

These results support two conclusions. First, T1 MRI contributes complementary information that improves performance over unimodal probes trained on the same matched cohort. Second, the strongest predictive signal in the current dataset comes from text, which explains why fusion remains text-dominant in both the quantitative and attributional analyses.

4.4 Multimodal Interpretability

Patient	Label	P(AD)	Vision (%)	Text (%)
671	AD	0.95	17.3	82.7
2058	AD	0.97	18.3	81.7
123	CN	0.21	12.0	88.0
934	CN	0.06	13.6	86.4
Average			**15.3**	**84.7**

Table 7: Per-modality attribution via Gradient×Input on the fusion MLP for four correctly classified test patients. All gradients are computed with respect to the predicted-class logit.

Modality Contribution Table 7 shows that the text modality dominates the fusion prediction across all four patients, contributing approximately 85% of the total Gradient×Input attribution. This finding is consistent with the stronger unimodal performance of the text encoder and reflects the rich diagnostic information encoded in structured clinical assessments. The neuroimaging modality contributes a consistent $\sim$15%, confirming that the brain scan provides complementary predictive signal.

Brain Spatial Importance Figure 6 presents the spatial importance maps derived from Gradient×Activation on the ViT's penultimate-layer patch tokens. The heatmaps reveal that the model focuses on ventricular and medial temporal regions, which is consistent with established AD neuropathology: ventricular enlargement due to brain atrophy and medial temporal lobe degeneration are among the earliest and most reliable structural markers of AD. AD patients (rows 1–2) exhibit more concentrated, focal activation patterns, while CN patients (rows 3–4) show diffuse spatial attention.

Text Token Importance Analysis of the text token attributions reveals that the fusion model assigns highest importance to numerical scores from cognitive assessments, particularly the Auditory Verbal Learning Test (RAVLT). For the two AD patients, the most important tokens correspond to low memory recall scores (3–8 words per trial), while for the CN patients, high recall scores (12–14 words) dominate the attribution. Additional important tokens include "Total" (aggregate test scores), "Animals" (Category Fluency Test), and "Time" (Trail Making Test). These findings align with clinical diagnostic practice, where cognitive testing-and RAVLT in particular-is among the most sensitive instruments for early AD detection.

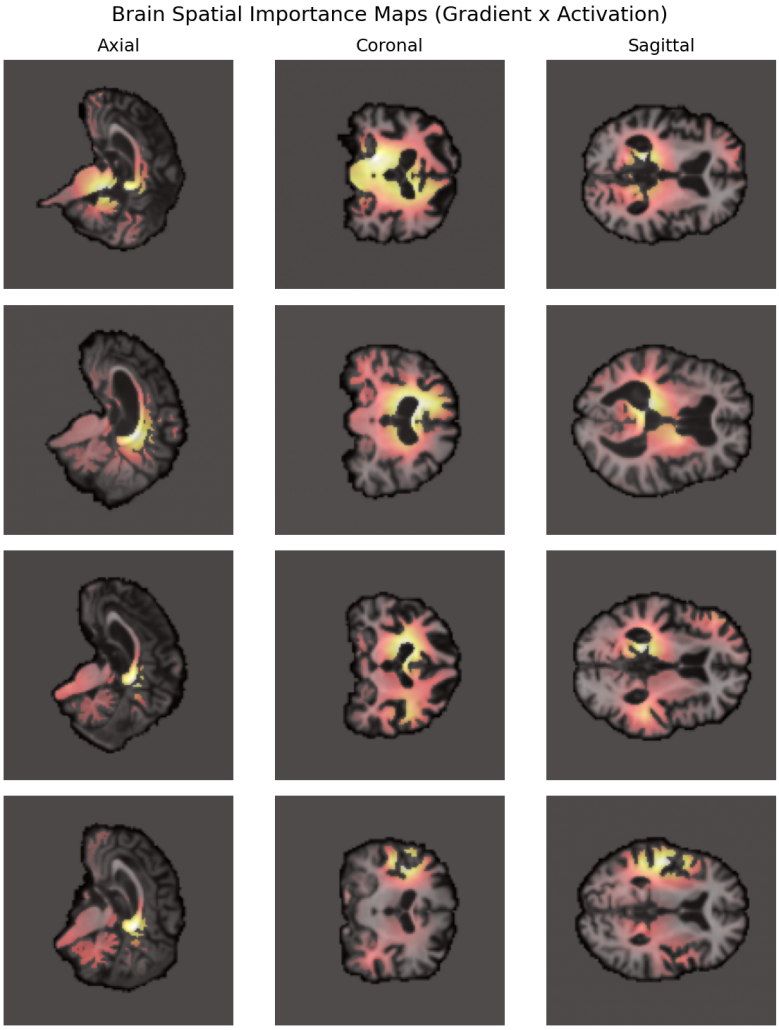


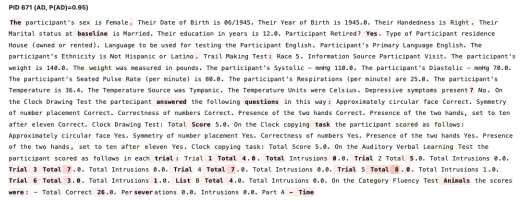
Figure 6: Brain spatial importance maps for four test patients (rows: 2 AD, 2 CN). Columns show axial, coronal, and sagittal mid-slices. Hot regions indicate high Gradient$\times$Activation importance from the fusion model's prediction. The model consistently highlights ventricular and medial temporal areas.

5 Discussion

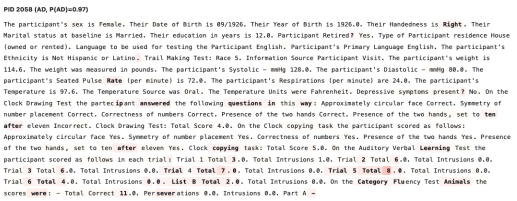
Our results demonstrate that late fusion of frozen pre-trained encoders is a viable strategy for multimodal AD classification, even under severe data constraints. With only 106 multimodal training subjects, the fusion model achieves the highest AUROC and balanced accuracy among all models trained under the same data budget, and approaches the performance of a text-only model trained on over $10\times$ more data (1142 subjects).

The dominance of the text modality ($\sim$85%) in the fusion prediction is clinically expected. The structured clinical notes contain direct cognitive assessment scores-particularly RAVLT, which is one of the most sensitive instruments for AD detection. These scores encode clinical information that is highly predictive of AD status. The neuroimaging modality contributes a consistent $\sim$15%, providing complementary signal that improves the balance between sensitivity and specificity relative to either unimodal model alone.

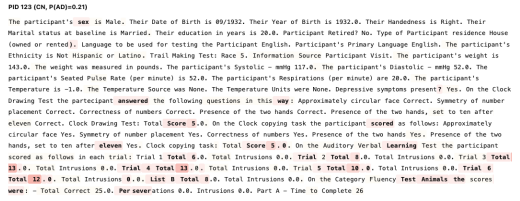
The interpretability analysis provides evidence that the model's predictions are grounded in clinically established biomarkers. The spatial importance maps highlight AD-relevant brain regions, while the



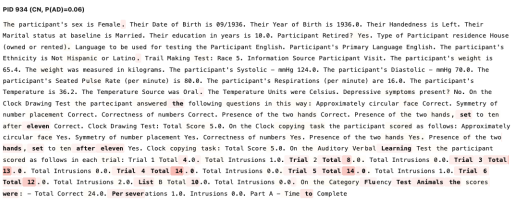
(a) PID 671 (AD, P(AD)=0.95)



(b) PID 2058 (AD, P(AD)=0.97)



(c) PID 123 (CN, P(AD)=0.21)



(d) PID 934 (CN, P(AD)=0.06)

Figure 7: Text token importance for four test patients. Highlighted tokens indicate high Gradient×Input attribution from the fusion model. AD patients (top) show high importance on low RAVLT recall scores (3–8), while CN patients (bottom) show importance on higher recall scores (12–14).

text token analysis identifies specific cognitive test scores that drive the prediction. This alignment with clinical knowledge increases confidence in the model's reasoning.

Limitations First, the dataset is small (214 patients, 106 for multimodal training), drawn from a single institution (ADNI), and lacks external validation. Second, the neuroimaging encoder was fine-tuned on only 84 T1 volumes, resulting in modest standalone performance (AUROC $= 0.64 \pm 0.05$) and consequently weaker visual features in the fusion model. Third, our late-fusion approach with frozen encoders is the simplest fusion strategy; cross-attention mechanisms or end-to-end training could potentially capture richer inter-modal interactions given sufficient data. Furthermore, the text modality consists of structured clinical assessments that directly encode diagnostic information; performance may differ substantially with unstructured clinical notes. We also recommend further investigation into whether the UMAP constraint systematically recovers correct attribution polarity.

6 Conclusion

Our main contribution is a multimodal framework for AD classification that combines T1 MRI with clinical text. Late fusion of frozen pre-trained encoders achieves a test AUROC of 0.820 ± 0.009 on 106 matched subjects. This is the highest AUROC and balanced accuracy among models trained on the same data. Text drives $\sim$85% of the fusion attribution, consistent with the diagnostic richness of structured cognitive assessments. Neuroimaging contributes $\sim$15% and improves specificity. Spatial importance maps and text token attributions align with established AD biomarkers, which supports the clinical grounding of the model's predictions.

On the text side, we reproduce and extend the monosemantic attribution framework of Mamalakis et al. [18]. TEO variants achieve RIS and ROS values orders of magnitude below all classical baselines. This confirms that multi-objective stability optimisation improves both sparsity and stability. The UMAP geometric constraint separates demographic trends from cognitive signals and recovers clinically consistent attribution.

These results show that attributions and multimodal approaches are useful tools for trustworthy clinical decision support.

7 Acknowledgements

The authors would like to thank Prof. Michail Mamalakis for his guidance in addition to principal lecturers Prof. Pietro Liò and Dr Tiago Azevedo.

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A Supplementary Material

A.1 Code Repository

The code is available at `https://github.com/LenteOrtu/neuro_interp_349`.

A.2 Team Member Contribution

We worked on this project horizontally. Each member initially worked independently to develop a stand-alone part, and we eventually unified the best working parts of each member's versions into our final solution. Umer's contribution centred around the text baseline and related attributional experiments. Oszkar initially worked on the neuroimaging unimodal baseline, before joining Fivos to develop the multimodal approach. Fivos worked on the neuroimaging baseline too, experimenting with several models, before choosing the final one from Oszkar's and Fivos's work. He worked on multimodal model and the multimodal interpretability.

All members contributed equally to the report write-up.

A.3 Further Experimental Results

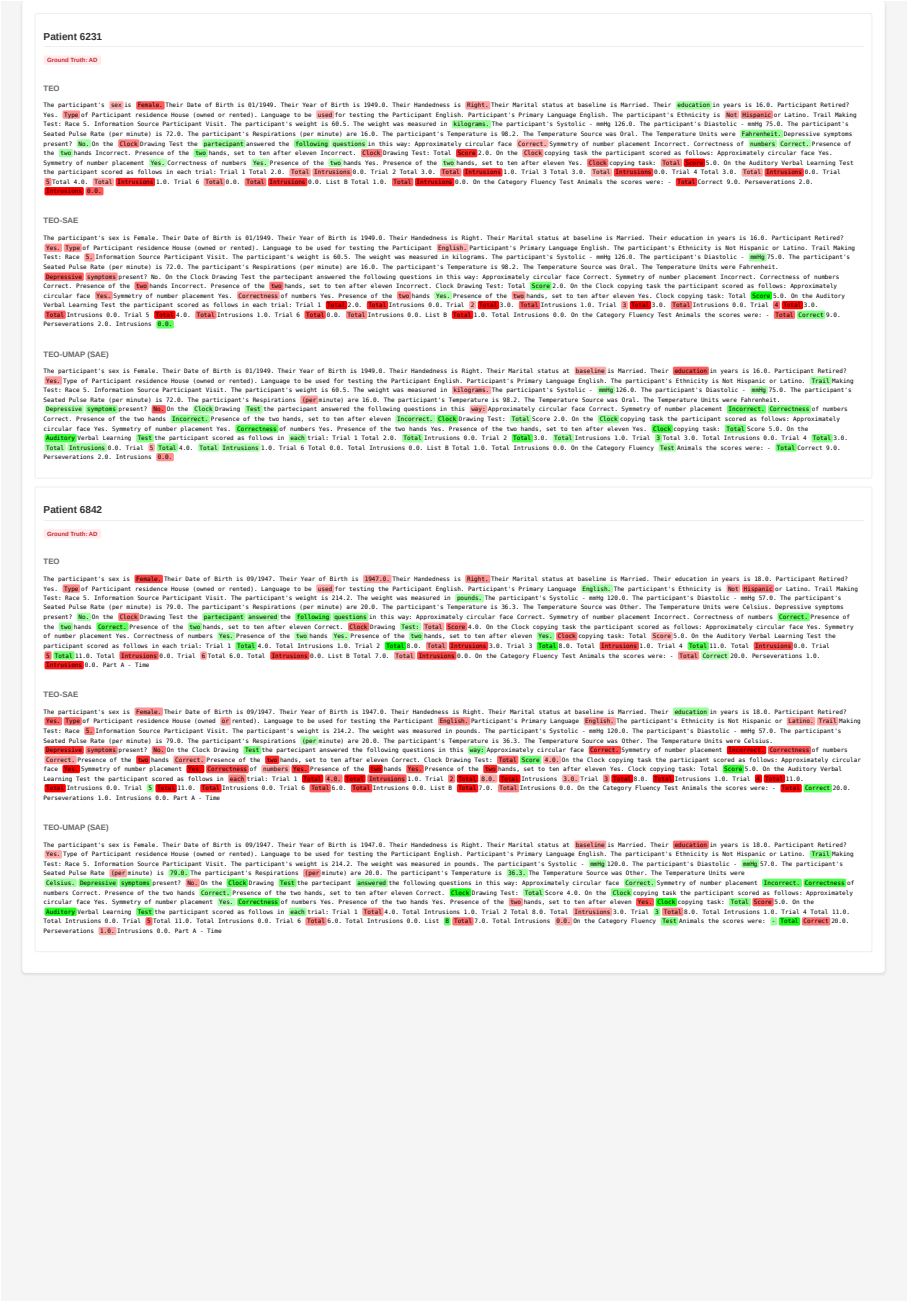
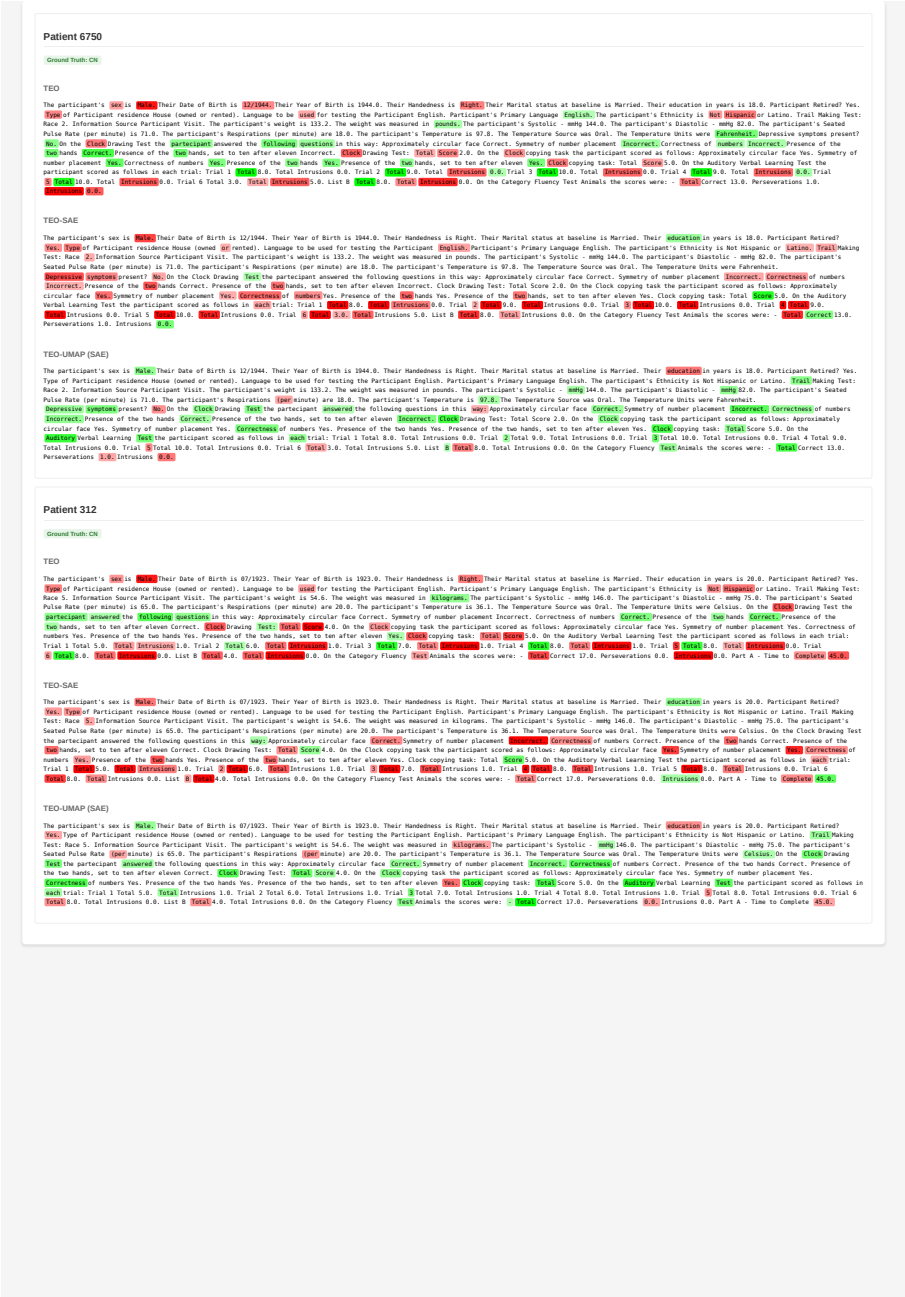


Figure 8: AD patient activations for PID 6231 and PID 6842



Patient 6750

Ground Truth: CN

TEO

The participant's sex is Male. Their Date of Birth is 12/1944. Their Year of Birth is 1944.0. Their Handedness is Right. Their Marital status at baseline is Married. Their education in years is 18.0. Participant Retired? Yes. Type of Participant residence House (owned or rented). Language to be used for testing the Participant English. Participant's Primary Language English. The participant's Ethnicity is Not Hispanic or Latino. Trail Making Test: Race 2. Information Source Participant Visit. The participant's weight is 133.2. The weight was measured in pounds. The participant's Systolic - mmHg 144.0. The participant's Diastolic - mmHg 82.0. The participant's Seated Pulse Rate (per minute) is 71.0. The participant's Respirations (per minute) are 18.0. The participant's Temperature is 97.8. The Temperature Source was Oral. The Temperature Units were Fahrenheit. Depressive symptoms present? No. On the Clock Drawing Test the partecipant answered the following questions in this way: Approximately circular face Correct. Symmetry of number placement Incorrect. Correctness of numbers Incorrect. Presence of the two hands Correct. Presence of the two hands, set to ten after eleven Incorrect. Clock Drawing Test: Total Score 2.0. On the Clock copying task the participant scored as follows: Approximately circular face Yes. Symmetry of number placement Yes. Correctness of numbers Yes. Presence of the two hands Yes. Presence of the two hands, set to ten after eleven Yes. Clock copying task: Total Score 5.0. On the Auditory Verbal Learning Test the participant scored as follows in each trial: Trial 1 Total 8.0. Total Intrusions 0.0. Trial 2 Total 9.0. Total Intrusions 0.0. Trial 3 Total 10.0. Total Intrusions 0.0. Trial 4 Total 9.0. Total Intrusions 0.0. Trial 5 Total 10.0. Total Intrusions 0.0. Trial 6 Total 3.0. Total Intrusions 5.0. List B Total 8.0. Total Intrusions 0.0. On the Category Fluency Test Animals the scores were: - Total Correct 13.0. Perseverations 1.0. Intrusions 0.0.

TEO-SAE

The participant's sex is Male. Their Date of Birth is 12/1944. Their Year of Birth is 1944.0. Their Handedness is Right. Their Marital status at baseline is Married. Their education in years is 18.0. Participant Retired? Yes. Type of Participant residence House (owned or rented). Language to be used for testing the Participant English. Participant's Primary Language English. The participant's Ethnicity is Not Hispanic or Latino. Trail Making Test: Race 2. Information Source Participant Visit. The participant's weight is 133.2. The weight was measured in pounds. The participant's Systolic - mmHg 144.0. The participant's Diastolic - mmHg 82.0. The participant's Seated Pulse Rate (per minute) is 71.0. The participant's Respirations (per minute) are 18.0. The participant's Temperature is 97.8. The Temperature Source was Oral. The Temperature Units were Fahrenheit. Depressive symptoms present? No. On the Clock Drawing Test the partecipant answered the following questions in this way: Approximately circular face Correct. Symmetry of number placement Incorrect. Correctness of numbers Incorrect. Presence of the two hands Correct. Presence of the two hands, set to ten after eleven Incorrect. Clock Drawing Test: Total Score 2.0. On the Clock copying task the participant scored as follows: Approximately circular face Yes. Symmetry of number placement Yes. Correctness of numbers Yes. Presence of the two hands Yes. Presence of the two hands, set to ten after eleven Yes. Clock copying task: Total Score 5.0. On the Auditory Verbal Learning Test the participant scored as follows in each trial: Trial 1 Total 4.0. Total Intrusions 0.0. Trial 2 Total 9.0. Total Intrusions 0.0. Trial 3 Total 10.0. Total Intrusions 0.0. Trial 4 Total 9.0. Total Intrusions 0.0. Trial 5 Total 10.0. Total Intrusions 0.0. Trial 6 Total 3.0. Total Intrusions 5.0. List B Total 8.0. Total Intrusions 0.0. On the Category Fluency Test Animals the scores were: - Total Correct 13.0. Perseverations 1.0. Intrusions 0.0.

TEO-UMAP (SAE)

The participant's sex is Male. Their Date of Birth is 12/1944. Their Year of Birth is 1944.0. Their Handedness is Right. Their Marital status at baseline is Married. Their education in years is 18.0. Participant Retired? Yes. Type of Participant residence House (owned or rented). Language to be used for testing the Participant English. Participant's Primary Language English. The participant's Ethnicity is Not Hispanic or Latino. Trail Making Test: Race 2. Information Source Participant Visit. The participant's weight is 133.2. The weight was measured in pounds. The participant's Systolic - mmHg 144.0. The participant's Diastolic - mmHg 82.0. The participant's Seated Pulse Rate (per minute) is 71.0. The participant's Respirations (per minute) are 18.0. The participant's Temperature is 97.8. The Temperature Source was Oral. The Temperature Units were Fahrenheit. Depressive symptoms present? No. On the Clock Drawing Test the partecipant answered the following questions in this way: Approximately circular face Correct. Symmetry of number placement Incorrect. Correctness of numbers Incorrect. Presence of the two hands Correct. Presence of the two hands, set to ten after eleven Incorrect. Clock Drawing Test: Total Score 2.0. On the Clock copying task the participant scored as follows: Approximately circular face Yes. Symmetry of number placement Yes. Correctness of numbers Yes. Presence of the two hands Yes. Presence of the two hands, set to ten after eleven Yes. Clock copying task: Total Score 5.0. On the Auditory Verbal Learning Test the participant scored as follows in each trial: Trial 1 Total 8.0. Total Intrusions 0.0. Trial 2 Total 9.0. Total Intrusions 0.0. Trial 3 Total 10.0. Total Intrusions 0.0. Trial 4 Total 9.0. Total Intrusions 0.0. Trial 5 Total 10.0. Total Intrusions 0.0. Trial 6 Total 3.0. Total Intrusions 5.0. List B Total 8.0. Total Intrusions 0.0. On the Category Fluency Test Animals the scores were: - Total Correct 13.0. Perseverations 1.0. Intrusions 0.0.

Patient 312

Ground Truth: CN

TEO

The participant's sex is Male. Their Date of Birth is 07/1923. Their Year of Birth is 1923.0. Their Handedness is Right. Their Marital status at baseline is Married. Their education in years is 20.0. Participant Retired? Yes. Type of Participant residence House (owned or rented). Language to be used for testing the Participant English. Participant's Primary Language English. The participant's Ethnicity is Not Hispanic or Latino. Trail Making Test: Race 5. Information Source Participant Visit. The participant's weight is 54.6. The weight was measured in kilograms. The participant's Systolic - mmHg 146.0. The participant's Diastolic - mmHg 75.0. The participant's Seated Pulse Rate (per minute) is 65.0. The participant's Respirations (per minute) are 20.0. The participant's Temperature is 36.1. The Temperature Source was Oral. The Temperature Units were Celsius. On the Clock Drawing Test the partecipant answered the following questions in this way: Approximately circular face Correct. Symmetry of number placement Incorrect. Correctness of numbers Correct. Presence of the two hands Correct. Presence of the two hands, set to ten after eleven Correct. Clock Drawing Test: Total Score 4.0. On the Clock copying task the participant scored as follows: Approximately circular face Yes. Symmetry of number placement Yes. Correctness of numbers Yes. Presence of the two hands Yes. Presence of the two hands, set to ten after eleven Yes. Clock copying task: Total Score 5.0. On the Auditory Verbal Learning Test the participant scored as follows in each trial: Trial 1 Total 5.0. Total Intrusions 1.0. Trial 2 Total 6.0. Total Intrusions 1.0. Trial 3 Total 7.0. Total Intrusions 1.0. Trial 4 Total 8.0. Total Intrusions 1.0. Trial 5 Total 8.0. Total Intrusions 0.0. Trial 6 Total 8.0. Total Intrusions 0.0. List B Total 4.0. Total Intrusions 0.0. On the Category Fluency Test Animals the scores were: - Total Correct 17.0. Perseverations 0.0. Intrusions 0.0. Part A - Time to Complete 45.0.

TEO-SAE

The participant's sex is Male. Their Date of Birth is 07/1923. Their Year of Birth is 1923.0. Their Handedness is Right. Their Marital status at baseline is Married. Their education in years is 20.0. Participant Retired? Yes. Type of Participant residence House (owned or rented). Language to be used for testing the Participant English. Participant's Primary Language English. The participant's Ethnicity is Not Hispanic or Latino. Trail Making Test: Race 5. Information Source Participant Visit. The participant's weight is 54.6. The weight was measured in kilograms. The participant's Systolic - mmHg 146.0. The participant's Diastolic - mmHg 75.0. The participant's Seated Pulse Rate (per minute) is 65.0. The participant's Respirations (per minute) are 20.0. The participant's Temperature is 36.1. The Temperature Source was Oral. The Temperature Units were Celsius. On the Clock Drawing Test the partecipant answered the following questions in this way: Approximately circular face Correct. Symmetry of number placement Incorrect. Correctness of numbers Correct. Presence of the two hands Correct. Presence of the two hands, set to ten after eleven Correct. Clock Drawing Test: Total Score 4.0. On the Clock copying task the participant scored as follows: Approximately circular face Yes. Symmetry of number placement Yes. Correctness of numbers Yes. Presence of the two hands Yes. Presence of the two hands, set to ten after eleven Yes. Clock copying task: Total Score 5.0. On the Auditory Verbal Learning Test the participant scored as follows in each trial: Trial 1 Total 5.0. Total Intrusions 1.0. Trial 2 Total 6.0. Total Intrusions 1.0. Trial 3 Total 7.0. Total Intrusions 1.0. Trial 4 Total 8.0. Total Intrusions 1.0. Trial 5 Total 8.0. Total Intrusions 0.0. Trial 6 Total 8.0. Total Intrusions 0.0. List B Total 4.0. Total Intrusions 0.0. On the Category Fluency Test Animals the scores were: - Total Correct 17.0. Perseverations 0.0. Intrusions 0.0. Part A - Time to Complete 45.0.

TEO-UMAP (SAE)

The participant's sex is Male. Their Date of Birth is 07/1923. Their Year of Birth is 1923.0. Their Handedness is Right. Their Marital status at baseline is Married. Their education in years is 20.0. Participant Retired? Yes. Type of Participant residence House (owned or rented). Language to be used for testing the Participant English. Participant's Primary Language English. The participant's Ethnicity is Not Hispanic or Latino. Trail Making Test: Race 5. Information Source Participant Visit. The participant's weight is 54.6. The weight was measured in kilograms. The participant's Systolic - mmHg 146.0. The participant's Diastolic - mmHg 75.0. The participant's Seated Pulse Rate (per minute) is 65.0. The participant's Respirations (per minute) are 20.0. The participant's Temperature is 36.1. The Temperature Source was Oral. The Temperature Units were Celsius. On the Clock Drawing Test the partecipant answered the following questions in this way: Approximately circular face Correct. Symmetry of number placement Incorrect. Correctness of numbers Correct. Presence of the two hands Correct. Presence of the two hands, set to ten after eleven Correct. Clock Drawing Test: Total Score 4.0. On the Clock copying task the participant scored as follows: Approximately circular face Yes. Symmetry of number placement Yes. Correctness of numbers Yes. Presence of the two hands Yes. Presence of the two hands, set to ten after eleven Yes. Clock copying task: Total Score 5.0. On the Auditory Verbal Learning Test the participant scored as follows in each trial: Trial 1 Total 5.0. Total Intrusions 1.0. Trial 2 Total 6.0. Total Intrusions 1.0. Trial 3 Total 7.0. Total Intrusions 1.0. Trial 4 Total 8.0. Total Intrusions 1.0. Trial 5 Total 8.0. Total Intrusions 0.0. Trial 6 Total 8.0. Total Intrusions 0.0. List B Total 4.0. Total Intrusions 0.0. On the Category Fluency Test Animals the scores were: - Total Correct 17.0. Perseverations 0.0. Intrusions 0.0. Part A - Time to Complete 45.0.

Figure 9: CN patient activations for PID 6750 and PID 312